steel is heated to the above critical temperature (from 750°C to 850°C), holding it at the temperature for a considerable period of the time and rapidly quenching it in water, oil or salt bath. Hardness is depending on the following factors,

1) Carbon content
2) Rate of cooling
3) Work size

Purpose of Hardening

1) To increase the hardness of the metal.
2) To resist wear and enable it to cut other metals.
3) To improve strength, elasticity, toughness and ductility.

TEMPERING

Commonly used in steel making, tempering is a heat treatment used to improve hardness and toughness in steel as well as to reduce brittleness. The tempering process creates a more ductile and stable structure. If the steel (or) any tool is very hard it will brittle. So the tempering operation is done to reduce the brittleness.

Process of Tempering

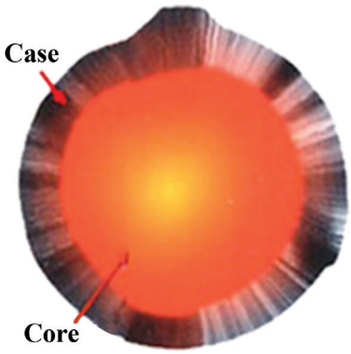
1. The steel is heat at below lower critical temperature after hardening.
2. Holding it for a considerable time.
3. Then cooling it slowly.

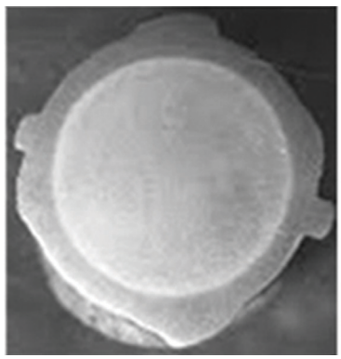
Purpose of Tempering

1. To decrease the brittleness of hardened steel.
2. To achieve the best mechanical properties in metal.
3. To stabilize the structure of metal.
4. To increase the toughness of steel.
5. To improve ductility.

CASE HARDENING

Case hardening is a process that is used to harden the outer surface metal while manufacturing a soft inner metal case. The case hardening process involves infusing addition carbon into the surface layer. In this process, chemical reaction made on the steel.







Case Hardening

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